

Productivity can nowadays be seen as the new currency of intellectual and social value. “Atomic Habits” by James Clear, “Getting Things Done” by David Allen, and many other books encourage people to seek out and build the best version of themselves, offering methods to improve performance, optimize their lives, and achieve their goals. In a rapidly changing world, productivity is therefore a decisive tool for achieving one’s dreams. However, it is striking to note that the rise of these life-optimization practices seems to coincide with an increase in certain mental and physical health concerns among the younger generation. Although this correlation does not imply a causal link, it is worth questioning the impact of perceived productivity on mental and physical health. In this research paper, we seek to determine: To what extent is there a systematic misalignment between perceived productivity and actual rest and well-being indicators?

1. Literature Review

Sleep and rest are widely recognized as fundamental determinants of cognitive performance, emotional regulation, and overall well-being. However, this essential need is often overlooked in favor of productivity, as people underestimate the amount of recovery time actually required. The rise in cases of mental and physical illness among young people, coupled with the constant evolution of productivity methods, has contributed to a growing interest in the relationship between perceived productivity, sleep behaviors, and well-being outcomes. Yet, the alignment between perceived productivity and actual rest indicators remains underexplored.

Several studies have examined the links between good sleep and cognitive performance. Overall, the literature is consistent: better sleep quality is associated with better productivity. In a study on the effect of sleep interventions on presenteeism (1) for example, the authors determined through experiments that cognitive behavioral therapy for insomnia (CBT-I) improved workers’ sleep quality, which led to fewer presenteeism issues.

Presenteeism is defined as a not so good productivity, despite being present at work. It can be considered as the illusion of productivity stemming from the perceived ratio of effort to work time. This confirms that there is an actual gap between perceived efforts and actual performance. Similarly, productivity depends on the quality of rest, not on perceived effort.

The paper notes, however, that these results were not the same for shift workers as they were for office workers. No particular improvement was noted for shift workers. The study did not offer any explanation for this, but this difference suggests that the work environment also plays a role.

Studies also suggest that quality is more important than quantity when it comes to sleep's restorative power. For example, a study experiment (2) showed that high-quality naps, taken under optimal conditions, have a greater positive impact on overall productivity than simply increasing the amount of sleep per night. However, it is possible that the results obtained for the group whose nights were extended would have been different if they had been placed under the same conditions as the group taking naps. Indeed, the body is designed to create an environment conducive to recovery in the evening, and it would have been interesting to examine how this factor affects the results. This reinforces the idea that the sleep quality that is to be considered for productivity, instead of perception of good sleep.

Furthermore, analysis of the available literature allows us to establish that the perception of exertion is a source of sleep debt awareness. A research paper on the impact of sleep quality on mental productivity (3) found that employees accumulate sleep debt - often without realizing it - and sacrifice sleep due to external pressure. Finally, a study on the relation between sleep and perceived productivity (4) brings us directly back to our topic of study. By comparing subjective and objective indicators of sleep quality and daytime alertness, the study concludes that there is a misalignment between perceived productivity and actual performance outcomes. Professionals who might perceive their long hours as productive were found to be 20% slower and made 30% more mistakes when their sleep quality was poor. This highlights a potential disconnect between perceived productivity and objective performance outcomes, reinforcing the need to examine how individuals assess their own rest needs.

However, existing studies focus mainly on the effects of sleep on productivity, but not on a possible misalignment between perceived productivity and actual rest and well-being indicators.

Additionally, the various studies focus primarily on the impact on work without establishing any link to daily functioning. For example, in the paper about sleep interventions and presenteeism (1), the researchers could have linked presenteeism to daily functioning, since improvements in focus and cognitive performance cannot be limited to work-related functions alone.

In terms of credibility, fairness, and objectivity, two of the referenced studies use indices such as the Pittsburgh Sleep Quality Index (PSQI) as well as neuropsychological tests (Stroop, Digit Span, Trail Making), which strengthens their technical credibility and objectivity. One of the referenced studies, itself based on other studies, estimates the risk of bias to be low to moderate. Nevertheless, their

critical analysis of the results allows us to retain this study as relevant documentation for our study.

This research aims to evaluate whether individuals' **perceived productivity accurately reflects their underlying rest and well-being state**, or whether systematic biases lead to miscalibration of rest needs.

2. Methodology

2.1. Research design and Hypothesis

For our study, we use a quantitative research approach to examine the relationships between academic performance and well-being indicators. Given the dataset we have - the Student Mental Health Dataset - we will conduct a proxy analysis to explore whether academic performance can serve as a reliable indicator of well-being. Indeed, our research question and literature review focus on actual rest, well-being indicators, and perceived productivity, but the dataset available to us does not directly capture these indicators. Therefore, this study employs a proxy-based approach, using academic performance as an indicator of productivity and mental health variables as indicators of well-being. This allows us to make an indirect assessment of potential misalignment between performance and underlying well-being.

While the existing literature emphasizes the role of sleep and perceived productivity in determining well-being outcomes, the dataset does not directly capture these variables. Consequently, this study adopts a proxy approach, using academic performance as an indicator of productivity and mental health variables as indicators of well-being. This allows for an indirect assessment of a potential discrepancy between academic performance and underlying well-being.

In this sense, the research question guiding this analysis is slightly reformulated. To answer the question of to what extent academic performance can serve as a reliable indicator of well-being, and whether there is a discrepancy between academic performance and mental health indicators, we proposed the following hypothesis.

H1: There is no positive relationship between academic performance and improved mental health outcomes among students.

H2: A significant proportion of high-achieving students exhibit symptoms of poor mental health (e.g., anxiety, depression, or panic attacks).

H3: There is no significant difference in mental health outcomes between high-achieving and lower-achieving students.

The selected analytical framework is applicable due to its capacity in determining patterns and contradictions in performance and well-being, thus answering the research question directly.

2.2. Data Collection

This study is based on the [Student Mental health](#), a public dataset containing anonymized data on students, their academic performance, and their mental health. As this is a secondary dataset, no primary data collection was conducted.

2.3. Sample & Selection Criteria

The selected dataset contains approximately 101 student responses. We retained only complete entries that included key variables such as GPA and mental health indicators.

2.4. Variables

Independent variable: academic performance, measured by GPA ranges

Dependent variables: these are the well-being indicators available in the dataset. They include:

- Depression (Yes/No)
- Anxiety (Yes/No)
- Panic attacks (Yes/No)

Derived Variables

Based on the variables we have, we create additional variables that will allow us to be more precise. First, we create a categorical variable that allows us to identify high-performing students—those whose GPA falls within the highest range. We also create a standardized variable for mental health risk indicators, which standardizes the information on whether or not the student reports one of these mental health conditions.

The use of GPA serves as a proxy for academic performance because it is a standard indicator used for this purpose. It is, in turn, used as a proxy for

perceived productivity. However, this variable does not accurately capture students' individual perceptions of productivity, which constitutes a limitation of this analysis.

3. Data Analysis Plan

Step 1: Descriptive analysis

The study begins with a descriptive analysis of the data. This involves, for example, assessing the distribution of GPA categories and mental health indicators.

Step 2: Group comparison

A derived variable allows us to classify students according to their GPA level (high, medium, low). At this stage, we analyze the prevalence of mental health concerns across the different groups.

Step 3: Misalignment analysis

Attention was specifically drawn to high-performing individuals who may have had mental health challenges, thus indicating a possible inconsistency between their performance and well-being.

Step 4: Comparative analysis

The mental health of high performers was evaluated against that of low performers to determine whether there was any difference in well-being between the two groups.

Comparisons among groups were used to uncover any possible discrepancies between academic achievement and mental well-being data. This strategy enables the identification of situations in which high academic performers demonstrate low well-being, signaling a disparity between academic achievement and mental well-being.

Limitations

This study provides an initial analysis of the potential misalignment between perceived productivity and effort and actual productivity and rest needs. However, a limitation of this initial analysis lies in the use of indirect indicators for certain variables, as direct measurements of these factors—namely productivity and rest—are not included in the dataset. Furthermore, the collection of subjective information on psychological well-being may have

introduced a bias in the results. Additionally, the sample size is relatively small, which can affect the robustness of the results.

4. Ethical Considerations

The data source that is used for this analysis is completely anonymised, without any personal identifiers, and is publicly available online. Therefore, no data privacy issue has been identified. Ethical issues should however always be considered when analyzing mental health data because of potential stigma that can arise out of the research. Another issue is self-reporting bias, especially in cases where some students self-assess their mental state.

In a broader context, which would address the initial research question, data anonymization remains essential. Collected data for this specific dataset must be based on objective, reliable measures and indicators (such as neuropsychological tests, the Pittsburgh Sleep Quality Index (PSQI), and the Epworth Sleepiness Scale (ESS)). Data collection includes the collection of subjective indicators (perceived productivity, perceived well-being, and rest indicators) and the collection of objective indicators (productivity and rest), all of which are anonymized.

This actually brings up a more deeper issue about data gathering. To get accurate measurements of the rest requirements and productivity levels of individuals, one would have to collect extremely personal information about individuals including their sleep patterns using a device such as a wearable tracker, their physiological status such as heart rate variation, and cognitive performance. This kind of data gathering can be considered highly invasive and may rise issues of privacy, informed consent, and data safety. Moreover, it may affect the behavior of participants making the study biased.

5. Attempted timeline

Literature review: Completed

Methodology design: Completed

Data analysis: - a day

Report writing: - a day

Final review and submission: -

References:

- (1) Takano, Y., Iwano, S., Aoki, S. et al. [A systematic review of the effect of sleep interventions on presenteeism](https://doi.org/10.1186/s13030-021-00224-z). BioPsychoSocial Med 15, 21 (2021). <https://doi.org/10.1186/s13030-021-00224-z>
- (2) NBER (2020). [*It's the quality of sleep that counts for boosting productivity*](#).
- (3) Ali Raza et al. (2024) "The Impact of Sleep Quality on Mental Productivity: A Comprehensive Analysis"
- (4) Sayre GM, Grandey AA, Almeida DM. Does sleep help or harm managers' perceived productivity? Trade-offs between affect and time as resources. J Occup Health Psychol. 2021 Apr;26(2):127-141. doi: 10.1037/ocp0000192. Epub 2020 Nov 5. PMID: 33151724; PMCID: PMC8352369.